

Hariharan Narlakanti

AI Engineer

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SUMMARY

Computer Science graduate (CGPA 8.89) with strong fundamentals in Data Structures & Algorithms, OOP, and DBMS, and hands-on experience building AI-powered backend systems using Python, FastAPI, Docker, and transformer-based NLP pipelines. Developed Retrieval-Augmented Generation (RAG) applications, semantic search systems, and LLM-powered inference services from prototype to production deployment.

EDUCATION

Malla Reddy University, Hyderabad

2022 – 2026

Bachelor of Technology – Computer Science & Engineering (AI & ML), CGPA: **8.89 / 10**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

AI/ML: Transformers, SentenceTransformers, PyTorch, ChromaDB, Groq API

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST API Design, JWT Authentication

Databases: PostgreSQL, MongoDB, MySQL

DevOps & Tools: Docker, Git, Linux, Render, Vercel

Frontend: React, Next.js, Vite

PROJECTS

Saaram — AI-Powered Telugu Summarization Platform

GitHub | Live Demo

- Designed and deployed a FastAPI-based AI inference service exposing **5+** REST endpoints, with adaptive model routing, fallback handling, and latency-aware optimization.
- Built and containerized a hybrid AI inference system (extractive and transformer-based summarization) using FastAPI and Docker, with lazy model loading under constrained compute resources.
- Benchmarked multiple summarization approaches (TF-IDF and transformer-based mT5 models) to optimize inference latency and select the most efficient configuration for production deployment.

ResearchCompass — AI Research Review Platform

GitHub | Live Demo

- Built a full-stack AI document intelligence platform (Next.js frontend, FastAPI backend, Dockerized) that parses uploaded PDFs and generates structured, section-cited AI-generated reviews.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and SentenceTransformer embeddings to ground LLM-generated critiques in specific paper sections and pages, with a provider-agnostic architecture currently served via the Groq API.
- Refactored an initial multi-agent workflow into a single streamlined AI reasoning pipeline, reducing orchestration complexity and improving output reliability.

Smart-FAQ — Full-Stack Semantic Search Platform

GitHub

- Built an AI-powered semantic search platform (FastAPI backend, React/Vite frontend, MongoDB) enabling natural-language FAQ retrieval via sentence-embedding similarity instead of keyword matching.
- Implemented JWT-protected admin APIs for FAQ management, with confidence-scored AI retrieval responses and fallback handling for low-confidence matches.
- Applied SentenceTransformer embeddings (all-MiniLM-L6-v2) for natural language understanding, computing semantic similarity between user queries and stored FAQs to return the best-matching answer with a confidence score.

ACHIEVEMENTS

- Best Project Award, Malla Reddy University – Telugu NLP capstone project.
- Co-authored a research paper on Telugu text summarization, submitted to CIS 2026 (Springer LNNS).